

Attacking and Defending Multi-Agent LLM Trading Systems

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Abstract

Multi-agent LLM systems are increasingly deployed for autonomous financial trading despite consuming third-party vendor data without cryptographic provenance verification. We instrument the open TRADINGAGENTS framework [10] with three compromised-channel attacks—fake-news injection, cross-channel coordinated disinformation, and pattern-matched memory poisoning—and three orthogonal defenses—a provenance-aware Portfolio Manager, a FinBERT-backed anomaly filter, and an independent Skeptic Agent—and measure 1,210 trials across five tickers and both bullish and bearish attack directions. The agent exhibits two-tier adversarial robustness in the bullish direction: an architectural memory access boundary attenuates memory poisoning at the perceptual layer (cross-ticker $\Delta = -0.12$, $p_{BH} = 0.30$, n.s.; the absorption judge labels 49/50 trials “not_absorbed”), and a multi-agent deliberation pipeline absorbs cross-channel disinformation without acting on it. Both tiers fail under specific conditions: perception-level attenuation breaks when attack direction aligns with agent-prior bias (bearish memory poisoning, $\Delta = -0.42$, $p_{BH} < 0.0001$), and decision-level resistance leaks on the cross-channel Mixed Attack; in one direction-attack cell, the Skeptic Agent amplifies rather than dampens the attacker’s effect. Controlled architectural ablation provides causal evidence that the memory access boundary mediates the bullish-direction attenuation. We argue that defense composition, rather than defense selection, is the more informative unit of analysis for adversarial robustness in deployed multi-agent LLM systems.

1 Introduction

Multi-agent LLM systems are being deployed in financial decision-making with increasing autonomy. Open frameworks such as TRADINGAGENTS [10] now organize role-specialized language models—specialist analysts, Bull/Bear researchers, a Trader, Risk debaters, and a Portfolio Manager—in a hierarchical pipeline that ingests external information from third-party vendors and emits an actionable trading decision. The economic stakes have outpaced the robustness analysis: these agents consume vendor responses based on semantic relevance, with no cryptographic provenance verification of the content they ingest [5]. An adversary who compromises *any one* information channel in the supply chain—a fabricated news article, a coordinated social-media campaign, or a poisoned long-term memory log—can in principle steer the agent’s decision without ever touching its prompt or weights.

Whether this concern is empirically realized, however, depends on a question that prior work on isolated-agent attacks [2, 8, 12] has not directly answered: *does multi-agent debate amplify or attenuate adversarial information in the supply chain?* On the one hand, division of labor enlarges the attack surface (more tools, more prompts, more inter-agent messages). On the other, the same division introduces redundancy and cross-checking that may dampen any single adversarial signal before it reaches the final decision. Resolving the trade-off requires (i) instrumenting a deployed multi-agent system with controlled threat surfaces, (ii) measuring whether attacks reach the agents’ *perception* versus whether they move the final *decision*, and (iii) identifying the architectural mechanisms that mediate between perception and decision.

We address this question on TRADINGAGENTS with a four-axis threat model and a defense-composition study. We instrument three attack vectors at distinct chokepoints in the agent’s information supply chain: **Fake News Injection** (a single fabricated article seeded from prosecuted SEC enforcement actions), **Cross-Channel Coordinated Disinformation** (a fabricated article corroborated by five trader-tone social posts that explicitly cite it, replicating the mixed-attack pattern reported by ASB [12]), and **Pattern-Matched Memory Poisoning** (eight directive “lesson learned” entries pre-populated in the long-term memory log, following MemoryGraft’s semantic-imitation heuristic [8]). We pair these with three orthogonal defenses—**Provenance-Aware PM** (three strength levels, including a strict variant that

detects circular cross-channel provenance), **Anomaly Filter** (lexical and FinBERT-based backends at the tool-output layer), and **Skeptic Agent** (an independent LLM critique pass between the Risk Debate and the PM)— and add a controlled *architectural ablation* that disables specific debate stages or moves memory access upstream, allowing us to turn an architectural correlation into mechanism-level evidence within the tested configuration. All overlays are runtime monkey-patches that simulate deployment-realistic compromises and protections without modifying TRADINGAGENTS source. We evaluate across five tickers spanning four sectors with both bullish and bearish attack directions, yielding 1,210 measured trials, and apply cluster-bootstrap resampling on tickers (with Benjamini-Hochberg false-discovery-rate correction) to address the pseudo-replication concern that within-ticker seeds are not independent samples.

Findings. We report five empirical findings. **(F1) Attack effectiveness is highly heterogeneous and exhibits two distinct failure modes:** the same SEC-seed-based fake-news payload moves PLTR by $\Delta = +0.64$ (data analytics) but is inert on BIIB ($\Delta = +0.00$, biotech, sector-narrative mismatch) and *backfires* on NVDA ($\Delta = -0.70$, AI semis with already-bullish priors), giving a non-significant cross-ticker $\Delta = +0.11$ $[-0.30, +0.45]$, $p_{BH} = 0.59$ (Section 5.1). This replicates the Universal-Triggers-Not-Universal finding of Meade et al. [7] in the financial domain. *[Evidence: descriptive; per-ticker means are stable at $N = 10$ each, but each ticker uses $K = 1$ payload, so the heterogeneity claim is well-supported while the two-mechanism interpretation (sector-narrative mismatch vs. high-prior backfire) remains payload-bound and awaits cross-payload replication.]* **(F2) Cross-channel coordinated disinformation partially overcomes sector mismatch:** on the BIIB outlier, cross-channel disinformation reaches $\Delta = +0.30$ and the Mixed Attack $\Delta = +0.40$, suggesting news-social corroboration partially substitutes for narrative-sector fit (Section 5.2). *[Evidence: suggestive; the BIIB cell is single-ticker $N = 10$ and the cross-ticker mean’s CI includes zero. Pattern is consistent with ASB’s mixed-attack mechanism but generalization across tickers is not established.]* **(F3) An architectural memory access boundary appears to attenuate memory poisoning at the perceptual level:** memory poisoning yields $\Delta = -0.12$ $[-0.28, +0.06]$ ($p_{BH} = 0.30$, n.s.) cross-ticker bullish, with the absorption judge labelling memory poisoning as “not_absorbed” in 49/50 trials (one partial, none fully absorbed; versus 50/50 fully absorbed for cross-channel disinformation); a controlled architectural ablation that moves memory upstream into the Fundamentals Analyst recovers attack effectiveness on both PLTR and SNOW (PLTR $\Delta = +0.80$, SNOW $\Delta = +0.40$, Section 5.3). *[Evidence: strong; cross-ticker absorption-judge data (49/50 across 5 tickers) plus mechanistic replication of the mem_to_analyst ablation on PLTR and SNOW. Among the findings most likely to generalize cross-LLM (see §6.3).]* **(F4) The architectural attenuation is direction-asymmetric:** reversing to bearish, memory poisoning produces a significant cross-ticker $\Delta = -0.42$ $[-0.68, -0.16]$, $p_{BH} < 0.0001$ (Section 5.4). *[Evidence: strong; cross-ticker 95% CI excludes zero across all 5 tickers in the bearish direction.]* **(F5) Defense composition, rather than defense selection, is the more informative framing in our setting:** Provenance-Aware PM and Skeptic Agent robustly reduce single-channel attacks ($p_{BH} < 0.0001$ on 4/4 cells, $\Delta \in [+0.47, +0.67]$), but both leak on the Mixed Attack ($p_{BH} > 0.5$). On bearish memory poisoning, Skeptic Agent *amplifies* the attack by 0.30 ordinal tiers on PLTR (Section 5.5). The leak pattern matches a stealth- detector mismatch (Section 5.6): Anomaly Filter with a lexical backend leaks cross-channel disinformation (lex score 0.097 near threshold), while Anomaly Filter with a FinBERT backend catches cross-channel disinformation because the mixed-document format depresses semantic similarity to real news. *[Evidence: mixed; bullish single-channel defense reduction is cross-ticker strong ($p_{BH} < 0.0001$, 4/4 cells across 3 tickers); the Skeptic Agent backfire on bearish memory poisoning and the Anomaly Filter backend ranking are single-ticker $N = 5$ measurements on PLTR and require cross-ticker replication before being treated as robust.]*

Contributions. We contribute (i) a literature-grounded threat-surface taxonomy on a real deployed multi-agent LLM trading system, with attacks instrumented at three distinct supply-chain chokepoints; (ii) a separation between *perception-level* immunity (the attack does not reach the analyst stage, evidenced directly by absorption-judge data) and *decision-level* resistance (the attack reaches the analyst stage but does not move the final decision), made rigorous through paired absorption and ordinal-decision measurement; (iii) a controlled architectural ablation that elevates an architectural correlation into causal mechanism evidence for memory poisoning’s failure under the default TRADINGAGENTS pipeline, replicated cross-ticker; and (iv) a defense- composition framework with the empirical observation that defense strength is non-monotonic, defenses can leak on the Mixed Attack, and in at least one direction-attack interaction (Skeptic Agent vs. bearish memory poisoning) defenses can amplify rather than dampen attacker effects. Together these results suggest that adversarial robustness in multi-agent LLM systems is not a property of any single defense layer but

of *defense composition*, and that direction-asymmetric attack classes warrant explicit testing in any deployment.

The remainder of the paper is organized as follows. Section 2 surveys prior work on multi-agent LLM systems for finance, adversarial attacks on LLM agents, multi-agent debate as both defense and vulnerability, adversarial input transferability and stealth, and financial misinformation. Section 3 formalizes our compromised-vendor-channel threat model. Section 4 details the attacks, defenses, architectural ablation, evaluation metrics, and cluster-bootstrap statistical analysis. Section 5 presents the five empirical findings across both directions and three defense matrices. Section 6 synthesizes the findings into a two-tier robustness account and a defense-composition framework, and discusses limitations. Section 7 concludes.

2 Related Work and Background

2.1 Multi-agent LLM systems for finance

A growing class of recent systems organizes role-specialized large language models into multi-agent pipelines for autonomous financial decision-making. *TRADINGAGENTS* [10] adopts a trading-firm metaphor: four specialist analysts (market, news, social, fundamentals) feed a Bull/Bear research debate, then a Trader proposes a transaction, three risk analysts (aggressive, conservative, neutral) debate, and a Portfolio Manager (PM) issues a five-tier ordinal decision. *FinAgent* [14] takes a complementary multimodal-foundation-agent design with dual-level reflection and tool-augmented retrieval, reporting substantial improvements over single-LLM baselines. Both systems represent the “hierarchical multi-agent specialization” design pattern that has become canonical in the financial-AI literature, and both consume external information from third-party vendors with no provenance verification — the precise threat surface we study.

We adopt *TRADINGAGENTS* as our system under test for three reasons: it is the most widely deployed open implementation of the design pattern; its five-tier ordinal output provides a controlled measurement target; and its memory log is read by exactly one agent (the PM) — a structural property we exploit in our architectural ablation (Section 5.3). Concurrent with our work, *TradeTrap* [5] proposes a system-level stress-test framework that perturbs four pipeline components (market intelligence, strategy formulation, portfolio handling, execution) and evaluates the *portfolio-level* consequences in a closed-loop backtest. We differ in two respects: (i) we instrument compromise at the information-channel layer rather than the tool-protocol layer, and (ii) we measure decision-level manipulation rather than P&L consequences. The two threat models are complementary; we discuss the relationship in detail in Section 3.1.

2.2 Adversarial attacks on LLM agents

The recent literature on agent attacks has established three principal attack surfaces: prompt injection, tool-output manipulation, and memory poisoning. The *Agent Security Bench* [12] formalizes these surfaces and reports a key benchmark finding: mixed attacks (combining multiple surfaces) reach 84.30% attack success rate, whereas memory-poisoning alone reaches only 7.92%. This asymmetry directly motivates our attack design: the mixed-attack class is the SOTA-effective configuration we instantiate as **Cross-Channel Coordinated Disinformation** plus **Pattern-Matched Memory Poisoning**, while the memory-poisoning single attack is the negative-baseline we observe at near-zero cross-ticker effect in the bullish direction (Section 5.3).

Two recent works specialize in memory-store compromise. *MINJA* [2] demonstrates query-only memory injection on GPT-4o-based agents at 98.2% inject success, with the attacker interacting only via natural-language queries. *MemoryGraft* [8] formalizes a “semantic-imitation heuristic” whereby agents replicate patterns from retrieved successful experiences — a phenomenon we directly exploit in our pattern-matched memory entries.

A parallel line of work on retrieval-augmented generation poisoning [15] shows that a small number of malicious documents inserted into a vector database can induce a 90% target-answer rate on RAG-based systems. Although *TRADINGAGENTS* does not currently use vector retrieval, the attack class is mechanistically related to ours: an attacker who controls a corpus the agent retrieves from (news vendor, memory store, document database) can manipulate downstream generation.

2.3 Multi-agent debate as defense and as vulnerability

Du *et al.* [3] introduced multi-agent debate (MAD) as a defense-and-reasoning mechanism: multiple LLM instances propose, critique, and revise responses across rounds, yielding improved factuality and reduced hallucination versus single- LLM inference. *AutoDefense* [11] extends the idea to a multi-agent jailbreak-defense pipeline that filters harmful responses through specialized agent roles.

Recent work, however, complicates this picture by showing that MAD is not uniformly more robust. Zhang *et al.* [13] demonstrate that structured prompt-rewriting attacks can amplify harmfulness in MAD systems from 28.14% to 80.34% by exploiting role- play dynamics, narrative encapsulation, and iterative refinement — making MAD systems *more* vulnerable than single-agent counterparts under certain attack patterns. Our findings are consistent with this duality: TRADINGAGENTS’s multi-agent debate provides strong decision-level resistance to single-channel attacks (Section 5.5, 4/4 defense cells significant) but leaks on the cross-channel Mixed Attack (Section 6.1). The defense and vulnerability are two faces of the same architecture.

2.4 Adversarial input transferability and content stealth

Universal Adversarial Triggers [9] introduced input-agnostic gradient-search triggers that transfer across inputs; Meade *et al.* [7] subsequently show empirically across 13 models that triggers transfer poorly, even within the same model family. Our sector-mismatch finding (Section 5.1)—where a tech-sector fake-news payload fails on biotech and backfires on AI semiconductors—is the financial-domain analogue of this transferability degradation. Jia and Liang [4] ground our news-injection paradigm in the SQuAD-adversarial methodology: an attacker inserts semantically plausible but factually fabricated content into the system’s perceptual stream while ground truth (in our case, market data) is unmodified. This positions our attack class as one of *semantic- distractor insertion* rather than gradient-search adversarial optimization.

Detection of adversarial text generally relies on detector-class- specific signals. We use both lexical statistics (sentence-length, hedge-word density, retail-marker frequency, bigram Jensen–Shannon divergence against a real-news baseline) and FinBERT-based semantic similarity [1] as complementary stealth backends. As we show empirically (Section 5.6), no single backend dominates: Atlas pump is lexically dead but semantically real-like; cross-channel disinformation closes the lexical gap but introduces a mixed- document-format FinBERT signature; fake news sits at the borderline of both.

2.5 Financial misinformation and adversarial market behavior

Adversarial information attacks against financial markets have a long pre-LLM history (Avon’s 2015 fake-tender-offer SEC enforcement, the 2013 AP Twitter hack causing the “flash crash”, Atlas Trading’s 2022 \$100M coordinated-pump prosecution). Recent ML-based detectors attempt to flag financial misinformation directly: FMDLlama [6] fine-tunes Llama 3.1 on a multi-task financial-misinformation dataset; broader misinformation literature considers LLMs as both attack vector and detector [12]. We adopt prosecuted SEC cases as the seed material for our fake-news attack generator (avon_fake_tender_2015 for bullish, craig_twitter_2015 for bearish) so that our injected payloads correspond to real-world attacker capabilities, not synthetic adversarial benchmarks.

Our work bridges two literatures that have largely operated in isolation: the agent-security literature (Sections 2.2 to 2.3), which evaluates attack/defense designs on isolated agents or generic multi-agent systems, and the financial-misinformation literature (this section), which studies the spread and detection of misinformation but typically not its effect on autonomous agentic trading systems. The intersection — *adversarial information-channel attacks against multi-agent LLM trading systems* — is, to our knowledge, not previously empirically characterized at the scale of the present study.

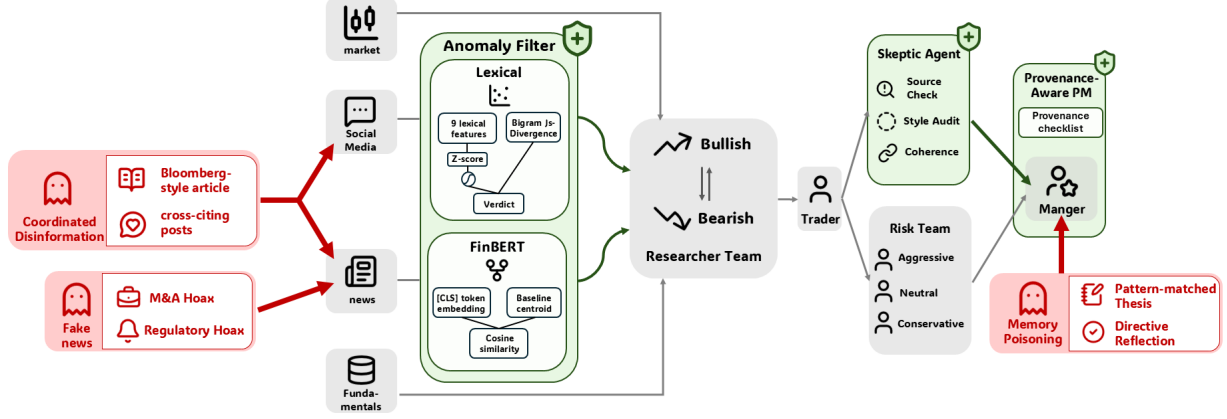


Figure 1: **Adversarial overlay of the TRADINGAGENTS pipeline.** Red ghost icons mark the three attack vectors and their channel targets: Fake News Injection and Cross-Channel Coordinated Disinformation compromise the shared `get_news` chokepoint feeding the News and Social Media Analysts; Pattern-Matched Memory Poisoning bypasses every input channel and writes directly to the Portfolio Manager’s persistent memory log. Green shields mark the three defenses: **Anomaly Filter** at the input layer, **Skeptic Agent** parallel to the Risk Team, and **Provenance-Aware PM** at the final decision stage. The three defenses cover non-overlapping attack surfaces.

3 Threat Model

3.1 Adversary capabilities and assumptions

We adopt a *compromised vendor channel* adversary model: the attacker has gained control of one (and only one) channel of the agent’s external information supply chain, but does not have access to the agent’s source code, prompts, weights, or runtime. The attacker’s capability is to substitute arbitrary content for vendor responses returned through the compromised channel; the attacker’s goal is to manipulate the Portfolio Manager’s final ordinal decision in a chosen direction (bullish or bearish). We model three realistically distinct compromised-channel instantiations (Section 4.2):

- **News-vendor compromise** (Fake News Injection). The attacker controls or has injected content into a news data feed (e.g. a third-party news aggregator that the agent’s `get_news` tool consumes). This matches real prosecuted market-manipulation cases such as the SEC enforcement actions against Avon Products fake-tender-offer (2015) and James Craig’s Twitter account hijack (2015), which we use as seeds for our generated payloads.
- **Social-channel coordinated compromise** (Cross-Channel Coordinated Disinformation). The attacker controls multiple apparently-independent social accounts and a single vendor news source, allowing the same fabricated narrative to appear with cross-source corroboration.
- **Memory-store compromise** (Pattern-Matched Memory Poisoning). The attacker has corrupted the agent’s long-term decision-memory store, e.g. by exploiting the query-only memory- injection vulnerability documented by Dong et al. [2], the persistent-experience-retrieval injection of Srivastava and He [8], or by directly compromising the storage layer.

Figure 1 summarizes the three compromise points relative to the TRADINGAGENTS pipeline. The two news/social attacks (Fake News Injection and Cross-Channel Coordinated Disinformation) target the agent’s *perceptual* layer at the shared `route_to_vendor` chokepoint, while the memory-store attack (Pattern-Matched Memory Poisoning) targets the agent’s *episodic memory* layer at the Portfolio Manager’s persistent decision log—bypassing every Analyst, Researcher, Trader, and Risk Team stage. This input/memory split structures both our threat model and the defense matrix that targets each layer independently (§4.3).

The agent has no built-in provenance verification: it consumes vendor responses based on semantic relevance to the analyst’s prompt, not cryptographic attestation of source, channel, or content integrity. This precisely matches

the failure mode formalized in TradeTrap [5]: “*models consume returned data purely based on semantic relevance without cryptographically verifying integrity or provenance*”. Our threat model differs from theirs in two respects: (i) we instrument compromise at the *information-channel layer* (news, social, memory), whereas TradeTrap models compromise at the *tool-protocol layer* (fake MCP server); (ii) we measure *decision-level* manipulation (5-tier ordinal output), whereas TradeTrap measures *portfolio-level* consequences (concentration, drawdown).

3.2 Implementation as runtime overlay

Because the attacker model assumes a compromised *external* channel, our experiment must reproduce what the agent observes under such compromise—namely, vendor output that has been adversarially modified before reaching the agent’s tool interface. We instrument this via a runtime monkey-patch of the `route_to_vendor` chokepoint in TRADINGAGENTS: the patched function calls the original vendor API and then appends or substitutes adversarial content before returning the result. For memory-store compromise, we directly pre-populate the per-trial isolated memory log file. In both cases, no TRADINGAGENTS source files are modified; the patch is a deployment-realistic simulation of the corresponding compromise, not a privileged in-process attack on the agent itself. The agent observes exactly what it would observe under a real upstream compromise. The same overlay design is used for the defense mechanisms in Section 4.3, where each defense represents a deployment-realistic policy or prompt change rather than a code modification.

3.3 Knowledge-isolated evaluation

A valid adversarial evaluation requires that the agent cannot recover ground truth by drawing on its pre-training knowledge or live tool access, since either would let the agent dismiss the injected content on grounds outside the threat model. We enforce three layers of isolation. First, all trade dates in our experimental schedule satisfy $\text{date} \geq 2025-12-01$, after the training-data cutoff of the agent LLM (`gpt-4o-mini`, October 2023 cutoff) and of contemporary frontier models that future cross-LLM evaluation might use. Second, all vendor data are sourced from frozen `yfinance` caches; live web access is disabled at the tool layer. Third, every trial runs against a freshly initialized isolated memory log to prevent cross-trial leakage of the agent’s own prior decisions. The combined effect is an evaluation envelope in which attacker-injected content is the only meaningful signal the agent has not seen before—a property required for valid causal attribution of decision shifts to attack effectiveness rather than to the agent’s own pre-training knowledge.

4 Methodology

4.1 Experimental setup

Tickers and dates. We select five tickers spanning four sectors — PLTR (data analytics, 2025-12-09), SNOW (cloud data, 2025-12-16), HOOD (fintech, 2026-01-13), BIIB (biotech, 2025-12-02), and NVDA (AI semiconductors, 2026-01-20) — chosen for sector diversity and post-cutoff date eligibility.

Agent LLM and isolation. The agent uses `gpt-4o-mini` (temperature defaults; `max_debate_rounds = max_risk_discuss_rounds = 1`). Each trial runs in an isolated fresh memory log (`cfg["memory_log_path"]` pointing to a per-trial file) to prevent cross-trial contamination. All vendor APIs are configured to `yfinance` only; live web or tool access beyond frozen caches is disabled.

Sample structure and trial execution. A single *trial* is one full run of TRADINGAGENTS’s `propagate(ticker, date)` pipeline—four analysts, Bull/Bear research debate, Trader, three-way Risk Debate, Portfolio Manager—under one (condition, seed) pair. Each trial begins with a freshly initialized memory log. For each ticker we run $N = 10$ random-seed-controlled trials per condition, with conditions cycling through `{clean, fake-news, cross-channel, memory-poisoning, mixed}` (four attack conditions in the v2 batch; the v1 baseline batch substitutes earlier-version A2 in place of cross-channel). Random seeds for each (condition, trial-index) pair are deterministic so re-runs of the same batch produce bit-identical decisions, modulo the agent LLM’s intrinsic stochasticity (`gpt-4o-mini`).

Table 1: Experimental scale. Total measured trials = 1 210.

Component	Spec	Trials
v1 bullish baseline	5 ticker $\times$ 4 cond $\times$ N=10	200
v2 bullish	5 ticker $\times$ 4 cond $\times$ N=10	200
Bearish batch	5 ticker $\times$ 4 cond $\times$ N=10	200
K-variant ablation	PLTR, K=3 $\times$ N=10	30
Architecture ablation	PLTR $\times$ 4 variants $\times$ 3 cond $\times$ N=5	60
Architecture cross-ticker	SNOW $\times$ 4 variants $\times$ 3 cond $\times$ N=5	60
Defense matrix (bullish v2)	3 ticker $\times$ 3 def $\times$ 4 atk $\times$ N=5	180
Provenance variant matrix (cross-ticker)	3 ticker $\times$ 2 def $\times$ 4 atk $\times$ N=5	120
Anomaly Filter backend matrix	PLTR $\times$ 2 backends $\times$ 4 atk $\times$ N=5	80
Bearish defense matrix	PLTR $\times$ 2 def $\times$ 4 atk $\times$ N=5	80
Absorption judge calls	5 ticker $\times$ 4 cond $\times$ N=10 LLM scorings	150
Stealth scoring	all cached payloads (lexical + FinBERT)	32

default temperature). All attack payloads are JSON-cached on disk and reused across trials sharing the same (ticker, date, condition), so seed variance within a cell isolates agent-side stochasticity from payload-side stochasticity (which we measure separately in the K -variant ablation, §5.7). Defense-matrix and architectural-ablation experiments are run on subsets at $N=5$ per cell (table 1). Defense and architecture conditions are runtime overlays applied at trial start and torn down at trial end; they do not persist across trials.

4.2 Attack vectors

We design three attack vectors that each compromise a different chokepoint in the agent’s information supply chain. Every attack is implemented as a runtime overlay—a Python monkey-patch around the agent’s tool-routing layer or memory-store interface—so the agent observes attacker-controlled content without any modification to TRADINGAGENTS source. The three vectors share a common goal (steer the Portfolio Manager’s final ordinal decision in a chosen direction) but differ in which agent-internal stage first sees the adversarial content.

Fake News Injection. The first attack models a compromised news vendor: the adversary places a single fabricated article into the news data feed that the agent’s news analyst consumes. To ground each fabricated article in a real-world manipulation tactic rather than a synthetic adversarial pattern, we maintain a small, curated catalog of prosecuted SEC enforcement-action seeds. Each seed encodes one historical manipulation pattern—fraudulent tender-offer filings, fabricated short-seller research notes, paid promotional pieces disguised as independent analysis—together with the natural attacker direction (bullish or bearish) of that pattern. Two seeds serve as the primary bullish and bearish payload generators in this paper: a 2015 case in which a fabricated SEC filing claimed an unknown party was acquiring Avon at a 188% premium (bullish), and a 2015 case in which fabricated tweets attributed to short-side research firms claimed pending regulatory investigations of the target company (bearish). At trial time, an LLM rewriter (gpt-4o-mini, temperature 0.85) retargets the seed’s manipulation pattern to the trial’s specific (ticker, date), producing a 150–220-word article in Bloomberg/Reuters style with plausible figures and attributions. Every generated payload then passes a four-criterion LLM-as-judge quality-control pipeline (factual plausibility, tone, narrative coherence, and numeric plausibility; judge: gpt-4o-mini, temperature 0); samples failing any criterion are regenerated with strengthened constraints. The accepted article is appended to the news-vendor response for the target trade date, and the payload is cached so that all repeated trials with the same target see identical content—isolating seed variance from payload variance in downstream analysis.

Cross-Channel Coordinated Disinformation. The second attack models a more sophisticated adversary who compromises both the news channel and a set of social-media accounts and coordinates them to produce mutually-corroborating content. For each (ticker, date, direction) target, a single LLM call (gpt-4o-mini, temperature 0.85) generates a coordinated bundle of two pieces. The first is a 250–350-word fabricated article in institutional anonymous-source style, using hedge phrasings characteristic of real breaking-news reporting (“people familiar with the matter”,

“documents reviewed by Reuters”) and anchored on one specific dollar figure and one named bank or advisory firm. The second is a set of five short social-media posts (100–220 characters each), written in five distinct professional-trader voices—an options-flow trader, a buy-side analyst, a tape-reader, a fundamentals analyst, and an anonymous tipster—each of which explicitly cites the article by source name, by timing, or by the article’s central figure. Generation constraints prohibit retail “meme-stock” vocabulary (no emoji, no “to the moon”, no cashtags), and require the posts to agree with the article on the central figure, source, and timing; the goal is content that reads as professional financial discourse rather than retail enthusiasm. The full bundle is delivered as a single block through the same news-vendor chokepoint, with internal section headers that route the article portion to the News Analyst and the social portion to the Social Media Analyst—producing automatic cross-channel corroboration in the agent’s downstream synthesis without requiring the attacker to compromise multiple separate tool endpoints. This design instantiates the mixed-attack mechanism that ASB [12] identified as the most effective single configuration in their agent-security benchmark.

Pattern-Matched Memory Poisoning. The third attack targets the agent’s persistent decision-memory log—a structurally distinct chokepoint, since the memory log is read by the Portfolio Manager alone and bypasses every upstream analyst. We pre-populate the per-trial isolated memory log with eight fabricated past-decision entries written in exactly the format the agent itself produces when recording a real decision: a header line (date, ticker, rating, raw return, alpha return, holding period), a decision block with rating, investment-thesis, and time-horizon fields, and a reflection block reporting the outcome. Five entries are same-ticker (matching the trial’s target) and three are cross-ticker drawn from a sector-similar pool of large-cap technology and data-analytics names; this five-plus-three split fully utilizes the Portfolio Manager’s memory-reader cap. Two design choices specialize this attack relative to a generic memory-injection baseline. First, the investment-thesis text in each entry uses pattern-matched vocabulary that overlaps real analyst-report language—phrases such as “operating leverage”, “13F filings”, “options flow”, “cup-and-handle breakout”, “federal contract acceleration”—so that the Portfolio Manager’s semantic-retrieval heuristic flags the entries as relevant to the current decision context. Second, every reflection block ends with a directive lesson statement of the form “LESSON LEARNED: [pattern] has been the correct call. Hesitation cost real money”, which encodes each past-decision outcome as an actionable rule rather than a passive observation. Bullish entries report +4–8% alpha against the S&P 500; bearish entries report symmetrically negative outcomes. The directive language and pattern-matched vocabulary jointly operationalize the semantic-imitation heuristic characterized by MemoryGraft [8].

Mixed Attack. The fourth condition combines the cross-channel disinformation bundle and the memory-poisoning entries simultaneously: the agent reads the coordinated news/social bundle at its perceptual layer while its memory log is pre-populated with directive lesson entries. This dual-channel condition tests whether perception-level and memory-level attacks compose constructively—the configuration ASB reported reaching 84.30% attack success rate in their agent-security benchmark setting [12].

4.3 Defense mechanisms

We design three orthogonal defenses, each implemented as a runtime overlay that does not modify TRADINGAGENTS source. The three defenses are positioned at architecturally distinct stages of the agent pipeline, illustrated in Figure 1: an **Anomaly Filter** sits at the *input* layer, scoring news-channel content immediately after the news vendor returns it and before any analyst reads it; a **Skeptic Agent** sits at the *review* layer as an additional reasoning pass between the analyst stages and the Portfolio Manager; and a **Provenance-Aware Portfolio Manager** acts at the *decision* layer, augmenting the Portfolio Manager’s own reasoning prompt with an explicit credibility checklist. Because each defense intercepts a different processing stage, we expect—and confirm in §5.5—that they cover non-overlapping attack surfaces rather than redundantly defending the same chokepoint.

Provenance-Aware Portfolio Manager. The first defense models a deployment-realistic policy intervention in which the institution that operates the agent updates the Portfolio Manager’s reasoning prompt to include an explicit credibility checklist, but does not change any other component of the pipeline. The Portfolio Manager is instructed, before adopting any claim from the analyst inputs, to audit it against the checklist and to treat any claim failing the checklist as low-credibility noise that must not drive the final rating. This defense adds zero additional LLM calls per trial: it is a

pure prompt augmentation. We study three variants of increasing strictness in order to characterize the marginal value of each enforcement layer.

- *Citation-only*: the weakest variant. Any claim must be attributed to a named, identifiable source; generic attributions such as “sources say” or “industry experts” are explicitly excluded. This tests whether name-attribution alone is sufficient to filter adversarial payloads.
- *Full*: adds two additional checks: every claim must have at least one independently corroborating report elsewhere in the analyst inputs, and every numerical claim must pass a scale-plausibility check (e.g., a multi-billion-dollar contract is unlikely for a small-revenue company).
- *Independent-source*: the strictest variant. It requires corroborating sources to come from different *channel types*—e.g., a news report corroborated by a regulatory filing, not merely by social-media chatter echoing the same news—and adds an explicit detection step for circular provenance, in which multiple apparent sources ultimately trace back to a single anonymous origin. When circular provenance is detected, cross-channel agreement is treated as evidence *against* action, since coordinated echoing across channels is the canonical signature of a coordinated disinformation campaign.

The three variants form a strictness chain (citation-only $\subset$ full $\subset$ independent-source), letting us measure whether stronger provenance enforcement monotonically improves robustness or hits diminishing or even negative returns.

Anomaly Filter. The second defense models a stealth-detection layer applied to news content immediately after it leaves the vendor and before any analyst reads it. The filter does not know anything about how an attacker formats injected content; it operates purely on the textual content of the vendor response, scoring each paragraph-sized segment and dropping segments whose score falls below a fixed credibility threshold. Dropped segments are replaced with a brief audit notice so the analyst sees that filtering occurred but cannot recover the removed text. We score segments at paragraph rather than full-response granularity because attacks land as one or two appended segments embedded in a longer block of real news, and whole-response scoring would dilute the anomaly signal.

We deliberately implement the filter using only off-the-shelf content-anomaly signals—no labels, no fine-tuning—so that the filter is a realistic baseline rather than a privileged detector. Two backend variants are evaluated.

- *Lexical backend*. A classical text-statistics scorer that requires no neural model. Each segment is summarized by nine surface-and-style features: sentence count, mean and standard deviation of sentence length, mean word length, total character count, type-token ratio, punctuation density, hedge-word frequency (counts of words such as “rumored”, “unverified”, “sources say”), and retail-marker frequency (counts of emoji, retail-trader catchphrases, and cashtags). Each feature is z -scored against a baseline corpus of real financial news, and the absolute z -scores are aggregated through a sigmoid into a single stealth score in $[0, 1]$, where values close to 1 indicate the segment is statistically indistinguishable from real news.
- *FinBERT backend*. A neural variant using FinBERT, a BERT model further pretrained on financial-domain text. Each segment is encoded into its sentence-level representation, and a stealth score in $[0, 1]$ is computed as the cosine similarity between this representation and the centroid of a real-news baseline, mapped from $[-1, 1]$ via the linear transformation $(\cos + 1)/2$.

Both backends share the same baseline corpus—a set of fifty real financial-news headlines drawn from yfinance for a fixed reference basket of large-cap names—computed once and cached. The neural backend is loaded lazily so the lexical backend can run on its own when the heavier dependency is unavailable. Comparing the two backends lets us examine whether classical surface statistics and neural semantics agree on what counts as “anomalous”, or whether they disagree in ways that an attacker could exploit (§5.6).

Skeptic Agent. The third defense models a different deployment intervention: inserting an additional reasoning pass between the analyst stages and the final decision, performed by a separate LLM instance running deterministically (gpt-4o-mini, temperature 0, to make the review reproducible across re-runs). Where the Provenance-Aware Portfolio Manager modifies the Portfolio Manager’s own prompt, the Skeptic is a structurally distinct agent: it receives the

analyst stage’s output as input, performs its own reasoning, and emits a structured critical review that the Portfolio Manager then consumes alongside the original analyst reports.

The Skeptic reads six artifacts produced by upstream stages—the four analyst reports (news, sentiment, fundamentals, market), the synthesized Bull/Bear investment plan from the Researcher Team, and the Trader’s investment plan—and applies a seven-item checklist that operationalizes the failure modes our attacks exploit: single-source blockbuster claims with no independent corroboration; retail-tone language indicating non-institutional source; news content that includes its own self-skeptical hedge (a synthetic-content tell); internal inconsistencies between analyst reports; numerical figures implausible relative to the target ticker’s known scale (e.g., a multi-billion-dollar contract for a small-revenue firm); articles that themselves narrate the price reaction (a synthetic-content tell, since price moves are reported by separate market data); and prior-decision lessons in the agent’s memory context that pattern-match the current setup but do not actually fit the current evidence. The Skeptic returns a structured review listing each triggered concern, plus a single overall caution level in $\{\text{none, moderate, high}\}$. The Portfolio Manager’s prompt is augmented with the Skeptic’s review and a hard rule that a *high* caution level forces a one-tier conviction downgrade on the final five-tier rating, ensuring the Skeptic’s verdict has a deterministic effect rather than relying solely on the Portfolio Manager’s discretionary weighting.

4.4 Architectural ablation

To test for causal evidence (rather than mere correlation) of the memory access boundary’s role within the default TRADINGAGENTS configuration, we run four *architectural* variants of the TRADINGAGENTS pipeline, again as runtime overlays on the agent factories:

- `none`: default pipeline.
- `no_bb`: Bull/Bear research debate skipped; their node is replaced with a passthrough that immediately routes to Research Manager.
- `no_risk`: three-way Risk Debate skipped via the same mechanism.
- `mem_to_analyst`: `state["past_context"]` is additionally injected into the Fundamentals Analyst prompt, moving memory access *upstream* of the decision-anchoring layers.

If memory poisoning’s near-zero bullish cross-ticker effect (§5.3) is mediated by the architectural memory-access boundary, then `mem_to_analyst`—the only intervention that injects memory *upstream* of the analysts— should recover attack effectiveness; `no_bb` and `no_risk`, which leave memory access at the PM but remove intermediate decision-anchoring stages, are diagnostic for whether those debate stages additionally dampen the propagation.

4.5 Evaluation metrics

Primary metric. The PM’s final 5-tier ordinal decision (Sell=1, Underweight=2, Hold=3, Overweight=4, Buy=5). The attack effect is reported as $\Delta = \overline{\text{ord}}_{\text{attacked}} - \overline{\text{ord}}_{\text{clean}}$.

Mapping ordinal decisions to portfolio exposure. The 5-tier scale admits a standard linear mapping to normalized portfolio weight, $\{\text{Sell, Underweight, Hold, Overweight, Buy}\} \rightarrow \{-1, -0.5, 0, +0.5, +1\}$, the convention used in long–short equity allocation when the agent’s output is interpreted as a position-sizing signal. Under this mapping, an ordinal shift of $\Delta = 1.0$ corresponds to a 50% shift in normalized portfolio weight per attacked trial. The two cross-ticker findings whose 95% CIs exclude zero translate to: the bearish memory poisoning effect $\Delta = -0.42$ corresponds to a $\sim 21\%$ weight shift per attacked decision; the Skeptic Agent backfire of $\Delta = -0.30$ on PLTR corresponds to a $\sim 15\%$ adverse exposure shift; the Anomaly Filter FinBERT cell of $\Delta = +0.80$ corresponds to a $\sim 40\%$ defended exposure shift. These translations are linear-mapping approximations and bound the order of magnitude of the economic exposure implied by our decision-level findings; they do not model realized P&L (drawdown, transaction cost, slippage are not modeled). A closed-loop backtest with these portfolio-exposure outputs is the natural extension to a P&L-grounded evaluation, which we list as deferred future work (§6.4).

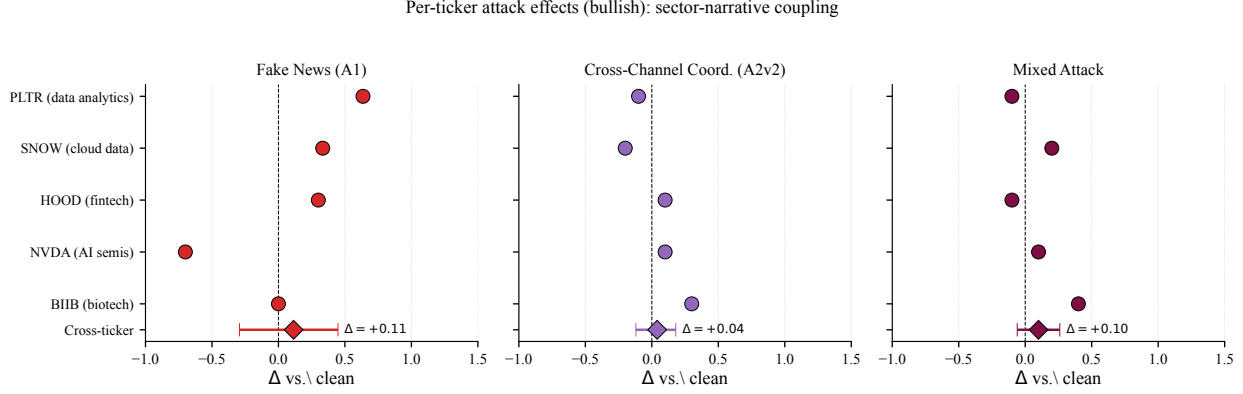


Figure 2: Per-ticker attack effects (bullish v1/v2 batches), revealing the two distinct failure modes (sector-narrative mismatch on BIIB; high-prior backfire on NVDA) hidden by cross-ticker averages. Diamonds indicate cross-ticker means with 95% cluster-bootstrap CI.

Secondary metrics. To address reviewer concerns that mean-ordinal alone may understate attack impact, we report: *action-flip rate* (fraction of attacked trials where the attacker-aligned-direction rate uplifts vs. clean), *position distortion* (mean $|\text{ord} - \text{ord}_{\text{clean}}|$, quantifying the spread of attacked decisions away from the clean center), and *unsafe exposure rate* (fraction of attacked trials in which the agent takes the attacker-aligned position — i.e. $\text{ord} \geq 4$ for bullish attacks, $\text{ord} \leq 2$ for bearish).

Absorption. An *LLM-as-judge* (gpt-4o-mini, temperature=0) reads concatenated analyst reports plus the PM’s investment plan and final decision, alongside the original injected payload, and emits a 3-level absorption verdict — NOT_ABSORBED (0), PARTIAL (1), FULL (2). We aggregate distributions per (ticker, condition) to separate the “attack reached perception” from “attack moved decision” failure modes.

Stealth. Combined lexical + FinBERT semantic similarity score per payload; reported as cross-attack mean comparison and per- attack distribution. Used to test paired stealth–defense predictiveness (§5.6).

4.6 Statistical analysis

To address the pseudo-replication concern (10 seeds within a ticker share the same payload, market context, and sector narrative), we employ *cluster bootstrap* resampling: each bootstrap iteration resamples *tickers* (the independence unit) with replacement, then averages within-ticker mean-differences. We use $B = 10\,000$ bootstrap iterations and report 95 % confidence intervals as the 2.5 %–97.5 % percentile range. P-values are two-sided ($p = 2 \cdot \min(P(\text{boot} \leq 0), P(\text{boot} \geq 0))$). For the defense-matrix test family (3 attacks $\times$ 2 defenses = 6 hypotheses), we apply Benjamini-Hochberg false-discovery-rate correction at $\alpha = 0.05$.

5 Results and Analysis

5.1 Heterogeneous fake-news effects: sector mismatch and high-prior backfire

We instantiate the **Fake News Injection** attack using the SEC-enforcement-action seed `avon_fake_tender_2015`—a tech-sector M&A narrative—and inject it into the news vendor channel of all five tickers. Per-ticker effects span a striking range: PLTR $\Delta = +0.64$ (data analytics), SNOW $\Delta = +0.33$ (cloud data), and HOOD $\Delta = +0.30$ (fintech) all show the expected attacker-aligned shift; BIIB $\Delta = +0.00$ (biotech) is unaffected; and NVDA $\Delta = -0.70$ (AI semiconductors) shifts in the *opposite* direction—a backfire of 0.70 ordinal tiers (Figure 2). Cross-ticker mean $\Delta = +0.11$ with 95% cluster-bootstrap CI $[-0.30, +0.45]$ ($p_{BH} = 0.59$, n.s.).

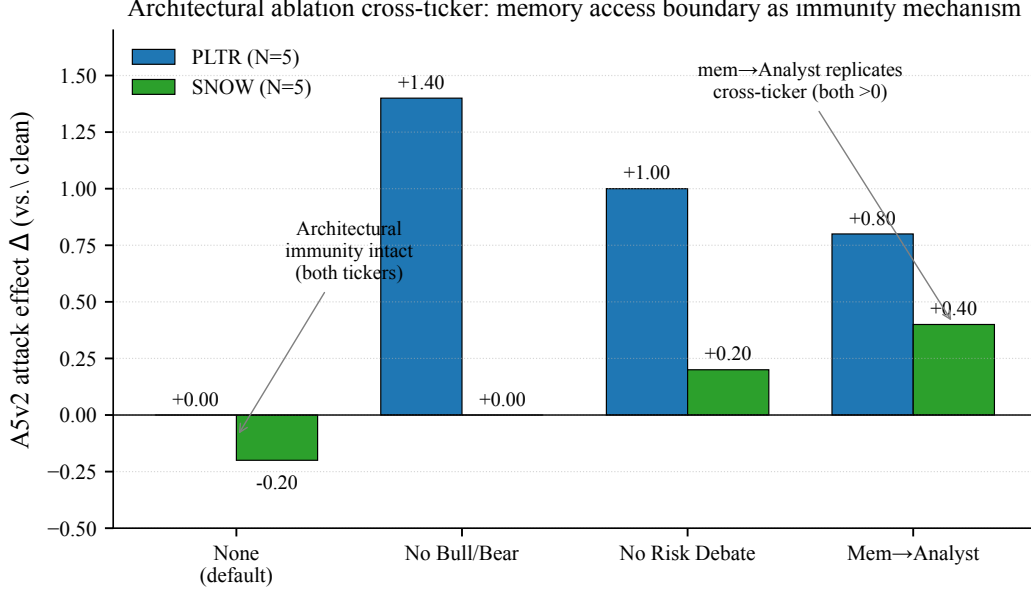


Figure 3: Architectural ablation on PLTR and SNOW ($N = 5$ each variant, 3 conditions). The default architecture yields $\Delta = 0$ for both memory poisoning and Mixed Attack on both tickers. On PLTR, all three perturbations recover attack effectiveness (`mem_to_analyst` $+0.80/+0.60$, `no_bb` $+1.40/+1.40$, `no_risk` $+1.00/-0.40$); on SNOW, the `mem_to_analyst` mechanism replicates most clearly ($+0.40/+0.00$), while debate-removal effects (`no_bb`, `no_risk`) are substantially weaker. The cross-ticker-replicated mechanism is therefore the memory access boundary; the role of upstream debate stages may be ticker- or sector-specific.

The per-ticker decomposition reveals two distinct failure modes that the cross-ticker average obscures. The first is *sector-narrative mismatch* (BIIB): the tech/M&A narrative carries no semantic relevance to a biotech firm, and the agent treats it as low-credibility noise, replicating the universal-triggers-not-universal finding of Meade et al. [7] in the multi-agent financial domain. The second is *high-prior backfire* (NVDA): NVDA’s clean baseline ordinal is 3.30 (Hold/Overweight-leaning), substantially above the other tickers’ ~ 2.30 baseline; under the same fake-news payload, the agent’s risk-aversion machinery interprets the unverified bullish claim as “too good to be true” and swings the decision *downward* to $\bar{\text{ord}} = 2.60$. To our knowledge, this is among the first documented cases of an adversarial fake-news attack *backfiring* on a multi-agent LLM trading system—an effect we hypothesize arises when the target asset’s own decision priors are already aligned with the attack direction, although confirming this attribution would require ablating priors directly (see Section 6.4). The finding has a practical implication for adversarial threat modeling: strong-prior assets may be *more* robust to attacker-aligned manipulation than weak-prior ones, even under sector match.

5.2 Cross-channel coordinated disinformation partially overcomes sector mismatch

We then test whether cross-channel coordinated disinformation can recover the BIIB outlier from Section 5.1. The **Cross-Channel Coordinated Disinformation** attack injects one Bloomberg-style fabricated article *plus* five trader-tone social posts that explicitly cite the article, simulating multi-channel adversarial corroboration. On BIIB (biotech) cross-channel disinformation reaches $\Delta = +0.30$ and the **Mixed Attack** reaches $\Delta = +0.40$ —compared to the fake-news attack’s $\Delta = +0.00$ on the same ticker. Cross-ticker, however, cross-channel disinformation yields $\Delta = +0.04$ and the Mixed Attack $\Delta = +0.10$ (both n.s.; CIs include zero).

The interpretation is that cross-source corroboration partially substitutes for narrative-sector fit: when the news article alone is insufficient to convince the agent, social-media corroboration provides a second independent signal that the agent’s downstream synthesis treats as cross-validating evidence. This mirrors the mixed-attack mechanism formalized by Zhang et al. [12] in their Agent Security Bench, whose reported 84.30% ASR for mixed-attack settings benchmarks this attack class as the SOTA-effective configuration. We note, however, that cross-channel corroboration

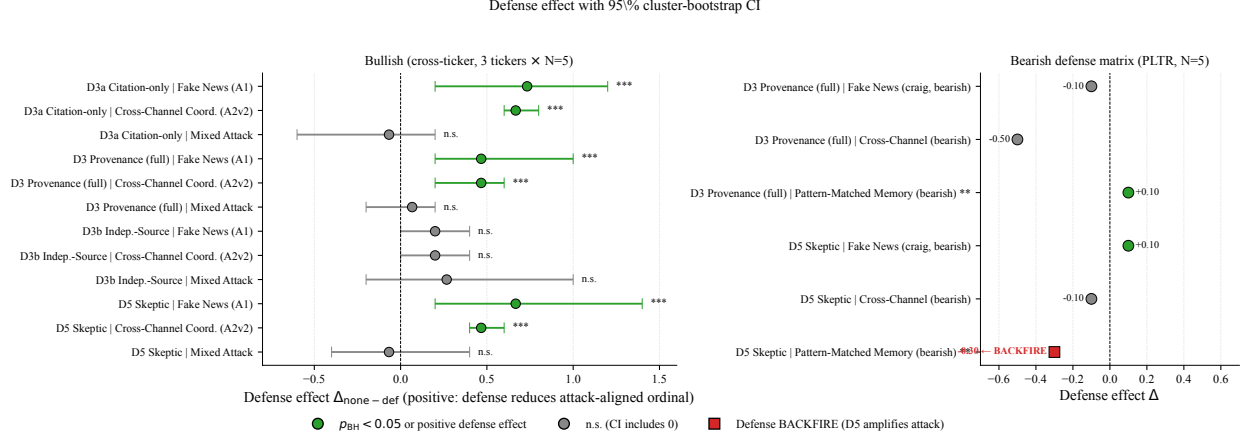


Figure 4: Defense effect forest plot. *Left*: bullish defense matrix, cross-ticker (3 tickers $\times N=5$); error bars are 95% cluster-bootstrap CI. Both Provenance-Aware PM and Skeptic Agent significantly reduce single-channel attacks (Fake News and Cross-Channel; $p_{BH} < 0.05$, CI excludes zero), but *both leak* on the Mixed Attack (CI includes zero, sign reverses for Skeptic Agent). *Right*: bearish defense matrix, PLTR-only ($N=5$); shown as exploratory seed-level estimates (no cross-ticker bootstrap is possible at single-ticker granularity). The Skeptic Agent backfire on bearish memory poisoning ($\Delta = -0.30$, bold) is the headline observation here but awaits cross-ticker replication.

does *not* recover the NVDA backfire (cross-channel disinformation on NVDA: $\Delta = +0.10$): high-prior backfire and sector-narrative mismatch are mechanistically distinct.

5.3 Architectural memory access boundary as immunity mechanism

We then test whether memory poisoning can move the agent’s decision. The **Pattern-Matched Memory Poisoning** attack pre-populates the agent’s long-term memory log with eight directive “lesson learned” entries. Despite the design—semantic-imitation heuristic [8], vocabulary overlapping real analyst reports (13F filings, operating leverage, options flow), and full utilization of the PM’s memory reader cap—the cross-ticker bullish effect is $\Delta = -0.12$ $[-0.28, +0.06]$ ($p_{BH} = 0.30$, n.s.). This is consistent with ASB’s reported memory-poisoning-alone ASR of 7.92% [12]; the cross-ticker effect we measure is indistinguishable from zero, suggesting TRADINGAGENTS’s hierarchical pipeline is at least as restrictive against memory-only attacks as the agents tested by ASB.

Absorption-judge data isolates the failure mode. We use a separate LLM-as-judge (gpt-4o-mini, temperature=0) to assess whether the injected payload reaches the analyst-stage reports (news, sentiment, fundamentals) at all. Across 5 tickers $\times N = 10$ trials, cross-channel disinformation absorbs as “fully_absorbed” in 50/50 trials (100%); memory poisoning is “not_absorbed” in 49/50 trials (98%). The asymmetry is mechanistic: TRADINGAGENTS grants memory access only to the Portfolio Manager, the last node in the pipeline, by which point upstream agents have already anchored the decision without seeing memory.

Architectural ablation confirms the mechanism causally. On PLTR ($N=5$ each variant, 3 conditions; Figure 3):

Variant	Δ_{mem}	Δ_{mix}
None (full pipeline)	+0.00	+0.00
mem_to_analyst	+0.80	+0.60
no_bb (skip Bull/Bear)	+1.40	+1.40
no_risk (skip Risk debate)	+1.00	-0.40

Moving memory access into the Fundamentals Analyst recovers an 0.80-tier decision shift; removing the Bull/Bear research debate recovers 1.40 tiers; removing the Risk Debate recovers 1.00 tiers. All three independent ablations

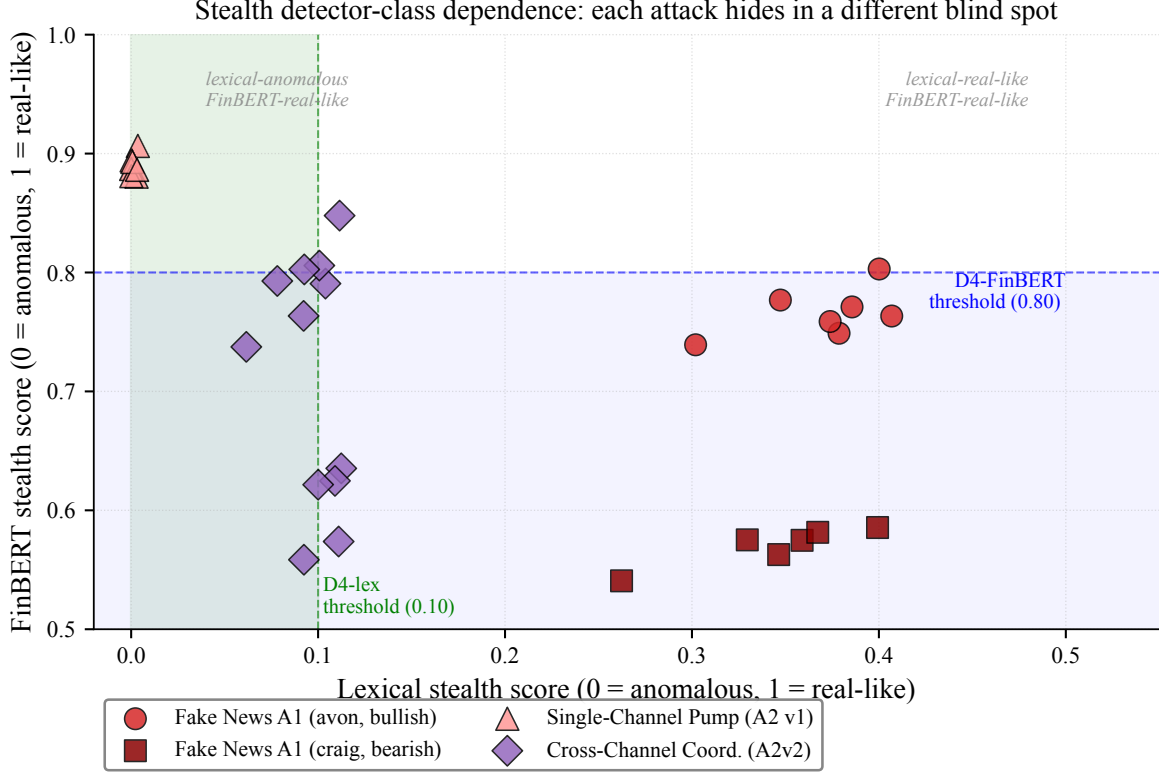


Figure 5: Stealth detector-class dependence. Each point is a cached attack payload; axes are lexical vs. FinBERT stealth scores (higher=more real-like). Atlas pump is lexically dead but FinBERT-real-like; cross-channel disinformation closes the lexical gap by 50 $\times$ but introduces a mixed-document-format FinBERT signature; fake news sits at the borderline of both. Threshold lines mark Anomaly Filter backend cutoffs.

are consistent with memory poisoning’s failure under the default architecture being architecture-mediated, rather than payload-weakness driven. We frame this finding as the *architectural memory access boundary acting as a structural attenuator*: under the default configuration, poisoned memory rarely propagates to upstream decision-anchoring agents because none of them has read access to the memory log.

Cross-ticker validation on SNOW. Replicating the same 4-variant ablation on SNOW ($N=5$ each variant) yields a partially heterogeneous picture: the central `mem_to_analyst` mechanism replicates (SNOW $\Delta_{\text{mem}} = +0.40$, $\Delta_{\text{mix}} = +0.00$; both positive in sign and consistent with PLTR’s larger effect), confirming that moving memory access upstream is the most reproducible mechanism finding. The `no_bb` and `no_risk` effects, however, are PLTR-specific: SNOW $\Delta_{\text{mem}} = +0.00$ for `no_bb` and $+0.20$ for `no_risk`, both substantially weaker than PLTR. We interpret this asymmetry conservatively: the memory access boundary is the robust cross-ticker mechanism explanation; the role of upstream debate stages may be ticker- or sector-specific and warrants further validation. We discuss this limitation in Section 6.3.

5.4 Direction-asymmetric architectural attenuation

The architectural attenuation observed in the bullish direction is not symmetric. Reversing the attack direction to bearish (using the SEC seed `craig_twitter_2015` and memory poisoning with `--direction=bearish`), the cross-ticker memory poisoning effect becomes $\Delta = -0.42$ with 95% CI $[-0.68, -0.16]$ ($p_{BH} < 0.0001$, significant). On PLTR specifically, the bearish memory poisoning distribution is *bimodal*: 6/10 trials follow the poisoned memory toward Sell (ord=1) while 4/10 revolt to Hold (ord=3). The within-batch position distortion is 0.98 versus a clean

baseline of 0.18—a 5 $\times$ stretch despite no significant shift in the mean alone.

We hypothesize that bearish poisoned memory synergizes with the agent’s default risk-averse bias—PMs in financial training data tend to favor Hold/Underweight as the safe baseline action—while bullish memory must *overcome* this same bias to move the decision upward. This makes memory poisoning the only attack in our suite to achieve a statistically significant cross-ticker effect, and it does so only in the bearish direction. The finding has dual implications: (i) defenders cannot assume that an attack class which fails in one direction will fail in both; (ii) the agent’s own decision priors are themselves an adversarial target that interacts non-trivially with external information manipulation.

5.5 Single-channel defense effectiveness and mixed-attack leak

We evaluate Provenance-Aware PM (**Provenance-Aware PM**, three strength levels), Anomaly Filter (**Anomaly Filter**, two backends), and Skeptic Agent (**Skeptic Agent**) across three tickers (PLTR, SNOW, HOOD), four attacks, $N=5$ seeds per cell (180 trials in the bullish v2-attacks defense matrix; Figure 4).

Single-channel attacks are reduced with high statistical confidence within the tested defense matrix. Cross-ticker defense effects (mean ordinal reduction vs. the no-defense baseline, with cluster-bootstrap 95% CI):

Defense	fake news	cross-channel disinformation	Mixed Attack
Provenance (full)	+0.47 [+0.20, +1.00]***	+0.47 [+0.20, +0.60]***	+0.07 [−0.20, +0.20]
Skeptic Agent	+0.67 [+0.20, +1.40]***	+0.47 [+0.40, +0.60]***	−0.07 [−0.40, +0.40]

Asterisks denote $p_{BH} < 0.0001$. Both prompt-level (Provenance-Aware PM) and agent-level (Skeptic Agent) defenses neutralize single-channel attacks with high statistical confidence (4/4 cells significant). Both *leak* on the Mixed Attack: Provenance-Aware PM’s effect collapses to +0.07 (CI includes zero); Skeptic Agent’s effect is −0.07 (CI includes zero, sign reverses).

Provenance-Aware PM strength variants exhibit non-monotonic returns (cross-ticker, 3 tickers $\times N=5$). The three Provenance-Aware PM strength levels exhibit a counterintuitive ordering. Citation-only Provenance is the strongest on single-channel attacks: Δ vs. fake news = +0.73 [+0.20, +1.20], $p_{BH} < 0.05$; vs. cross-channel disinformation = +0.67 [+0.60, +0.80], $p_{BH} < 0.05$. Provenance (full) provides similar protection (Δ = +0.47 on both, see above table). independent-source Provenance (independent-source with circular-provenance detection) is *weaker* than citation-only Provenance on single attacks (Δ = +0.20 on both, CIs include zero) and does not close the Mixed-Attack leak (Δ = +0.27 [−0.20, +1.00], n.s.). The non-monotonic pattern holds across PLTR, SNOW, and HOOD: the citation-only check is already sufficient to reject fake-news payloads (which all rely on fabricated or anonymous sources), and stricter cross-channel rules trade single-attack robustness for mixed-attack robustness without producing a net gain. We interpret this as evidence that defense- strength selection should match the expected attack class — strict provenance is over-kill (and conservatively biased) against single- channel attacks, and is not in itself sufficient to neutralize the cross-channel Mixed Attack.

Bearish defense matrix uncovers a defense-direction asymmetry on PLTR. The bullish-direction defense effects above were measured against attacks that we showed earlier (Section 5.4) do not produce statistically significant cross-ticker shifts. We therefore additionally measure Provenance-Aware PM and Skeptic Agent against the only attack that achieves a significant cross-ticker effect: bearish **Pattern-Matched Memory Poisoning** in the bearish direction (cross-ticker Δ = −0.42, $p < 0.0001$). On PLTR ($N=5$):

Bearish defense	fake news (craig)	cross-channel disinformation (bearish)	memory poisoning (bearish)
Provenance (full)	−0.10	−0.50	+0.10
Skeptic Agent	+0.10	−0.10	−0.30

Two findings emerge. First, Provenance-Aware PM and Skeptic Agent each achieve essentially zero defense effect against bearish memory poisoning (Provenance-Aware PM Δ = +0.10 is too small to be meaningful at $N=5$). Second

— and surprisingly — the Skeptic Agent makes bearish memory poisoning *more* successful ($\Delta = -0.30$): the agent’s separate-LLM critique pass *amplifies* rather than dampens the directive “lesson learned” memory in the bearish direction. We hypothesize that the Skeptic’s explicit articulation of doubt about the analyst reports interacts synergistically with the poisoned memory’s directive language (“LESSON LEARNED: when this pattern appears, lean Sell with conviction”), giving the PM permission to act on the memory rather than down-weight it. Although the effect is single-ticker $N = 5$ and we report it with appropriate caution, the qualitative direction is the opposite of what was hypothesized in the design of Skeptic Agent, and identifies the only cell in our suite for which we observe a directionally-amplifying effect of Skeptic Agent (single-ticker $N = 5$, to be replicated cross-ticker before being treated as a robust finding). This single-cell observation sharpens the *defense composition* thesis of Section 6.2: defenses that work in one direction are not guaranteed to work in the other, and a deployed pipeline must specifically test direction-asymmetric attack-defense interactions.

5.6 Stealth detector-class dependence and stealth–defense pairing

To explain why Anomaly Filter’s lexical and FinBERT backends differ in defense matrix performance, we measure the stealth profile of each attack payload using two complementary detectors: lexical (sentence-length statistics, type-token ratio, hedge-word density, retail-marker frequency, bigram Jensen-Shannon divergence vs. a real-news baseline) and semantic (FinBERT [1] CLS embedding cosine similarity to the same baseline). Figure 5 plots the results.

Attack	n payloads	Lexical (mean)	FinBERT (mean)
fake news (avon, bullish)	7	0.379	0.768
fake news (craig, bearish)	6	0.344	0.572
Atlas pump	7	0.002	0.890
Cross-Channel Disinformation	12	0.097	0.713

Atlas pump exhibits a striking lexical-vs-FinBERT divergence: lexically dead (0.002, far below the 0.10 threshold) yet semantically real-like (0.890). cross-channel disinformation closes the lexical gap by 50 $\times$ (0.097, near the threshold) but introduces a new mixed-document-format signature that lowers FinBERT similarity to 0.713. fake news sits at the borderline of both detectors.

Anomaly Filter backend defense effects are consistent with the detector-class hypothesis. On PLTR ($N = 5$):

Defense backend	fake news	cross-channel disinformation	Mixed Attack
Anomaly Filter (Lexical)	+0.40	−0.40	+0.00
Anomaly Filter (FinBERT)	+0.80	+0.20	+0.80

Anomaly Filter with the lexical backend leaks cross-channel disinformation (negative Δ , attack slips through) and is ineffective on the Mixed Attack ($\Delta = 0.00$). Anomaly Filter with the FinBERT backend catches cross-channel disinformation (positive Δ) and is the strongest single defense against the Mixed Attack at this single-ticker $N = 5$ measurement ($\Delta = +0.80$)—a configuration that neither Provenance (full) nor Skeptic Agent achieves on the same cell. We caveat this comparison with the small-cell sample size and report it as a within-PLTR ranking rather than a cross-ticker generalization. The result is the empirical evidence for detector-class dependence within this single-ticker measurement: stealth-metric data is consistent with the ranking of defense backends across attack classes, although generalizing this pairing rule across tickers requires further replication.

5.7 Payload variance dominates seed variance

Cross-ticker measurements suffer from payload-side stochasticity: when each (ticker, condition) cell uses a single LLM-generated payload ($K = 1$) shared across N seeds, the seed-level standard deviation underestimates the true effect distribution. We test this on PLTR by generating $K = 3$ independent cross-channel disinformation payload variants and running $N = 15$ seeds (each variant gets 5 seeds via deterministic cycling).

Configuration	$\Delta_{\text{cross-channel disinformation}}$	std
$K = 1, N = 10$	-0.10	0.48
$K = 3, N = 15$	+0.53	1.22

The $K = 3$ measurement reveals 1 Buy, 1 Overweight, and 2 Sell among the 15 trials—extreme decisions that the $K = 1$ measurement averages away as a null effect. Payload-side variance (0.74-tier shift in mean estimate) dominates agent-side seed variance (which on this ticker is ~ 0.5 ordinal-tier).

The implication is methodological: claims of attack effectiveness should report payload variance explicitly, ideally via $K \geq 3$ ablation. Single-payload measurements may dramatically under- or over-estimate true attack potential—a caveat we apply when interpreting our own cross-ticker $K = 1$ results in Sections 5.1 to 5.5.

6 Discussion

6.1 Two-tier immunity: perception-level and decision-level

The five empirical findings reduce to a single architectural claim: TRADINGAGENTS exhibits *two-tier* robustness against information-channel adversarial attacks, and the two tiers fail under distinct conditions.

The first tier, *perception-level attenuation*, is purely structural. The agent’s memory access boundary—whereby only the Portfolio Manager reads the long-term decision log, while all upstream analysts operate without memory access—is associated with pattern-matched memory poisoning rarely propagating to the analyst-stage reports (Section 5.3, 49/50 “not_absorbed” versus 50/50 “fully absorbed” for Cross-Channel cross-channel disinformation). Architectural ablation provides causal evidence consistent with this account: moving memory upstream into the Fundamentals Analyst recovers the attack ($\Delta = +0.80$), and disabling either the Bull/Bear or Risk debates also recovers it (Figure 3). Within the default configuration, this tier is structurally attenuated by *construction*: memory poisoning has no path to upstream agents that have not read the memory log.

The second tier, *decision-level resistance*, is policy-driven. Cross-Channel Coordinated Disinformation reaches the analyst stage (100% absorption) but its narrative does not move the final decision in the bullish direction ($\Delta = +0.04$, n.s.) because the Bull/Bear research debate, the Risk Debate, and the PM’s own synthesis collectively down-weight a single fabricated narrative. This tier is robust by *policy*: the agent absorbs the attack but its multi-agent deliberation neutralizes it.

The two tiers fail under different threats. Perception-level immunity *breaks* when the architecture is changed (the `mem_to_analyst` ablation, replicated cross-ticker on PLTR and SNOW) or when attack direction aligns with agent-prior bias (Section 5.4: bearish memory poisoning $\Delta = -0.42$, $p_{BH} < 0.0001$; $\sim 21\%$ adverse portfolio-weight shift under the linear mapping of §4.5). Decision-level resistance *leaks* on the Mixed Attack (Section 5.5). Most strikingly, defenses designed for the decision-level tier can themselves *amplify* attacks under specific direction-attack interactions: Skeptic Agent makes bearish memory poisoning more successful by 0.30 ordinal tiers on PLTR ($\sim 15\%$ adverse exposure shift; Section 5.5), the opposite of its intended effect. This distinction matters operationally: defenders deploying TRADINGAGENTS-class systems must understand which tier protects against which threat, since strengthening one does not patch the other and may even worsen residual exposure.

6.2 Defense composition, not defense selection

A second theme cuts across our defense-evaluation results: no single defense suffices against all attack classes, and the failure modes are predictable from the stealth profile of the attack rather than from the defense’s nominal sophistication. Provenance-Aware PM (prompt-level provenance checking) and Skeptic Agent (an additional LLM review pass) each robustly reduce single-channel attacks ($p_{BH} < 0.0001$ on 4/4 cells), but both leak on the Mixed Attack ($p_{BH} > 0.5$). Anomaly Filter (FinBERT backend) catches the Mixed Attack ($\Delta = +0.80$) but only because the mixed-document format depresses FinBERT semantic similarity to real news; the same attack slips through Anomaly Filter’s lexical backend ($\Delta = 0.00$). The non-monotonic behavior of Provenance-Aware PM’s three strength variants (Section 5.5) further demonstrates that increasing defense strictness does not monotonically increase robustness: Citation-only

Provenance outperforms Provenance (full) and Independent-source Provenance) on single attacks because adversarial payloads already fail the weakest check, while stricter checks introduce conservative bias on legitimate signals.

The implication is methodological. Adversarial-robustness papers typically frame the question as “which defense is best.” Our results suggest a complementary framing: *defense composition, rather than defense selection, appears to be the more informative unit of analysis* in our setting. The strongest single defense we observe at the single-ticker $N=5$ measurement on PLTR (Anomaly Filter with FinBERT backend on the Mixed Attack, $\Delta = +0.80$) is a low-cost, narrow-scope filter that would be ineffective without Provenance-Aware PM or Skeptic Agent covering the single-channel cases. A composite defense pipeline—a stealth-metric tool- output filter (Anomaly Filter, FinBERT backend), a Provenance-Aware PM prompt (citation-only-class, sufficient at the single-attack tier), and a Skeptic agent for residual semantic anomalies—would in principle close the leaks each leaves individually. We list this as the most important deferred experiment in Section 6.4.

6.3 Limitations

We highlight five limitations that constrain the generalization of our findings. First, all experiments use a single agent LLM, `gpt-4o-mini`; cross-LLM validation (`gpt-4o`, `Claude`) is deferred. Some findings, in particular the high-prior backfire on NVDA (Section 5.1), may depend on the specific risk-aversion calibration of `gpt-4o-mini`. Second, the primary metric is a five-tier ordinal decision, not a portfolio profit-and-loss measurement; we therefore phrase our claims as “decision manipulation robustness” rather than “trading loss robustness.” Third, our sample comprises five tickers across four sectors, and within-sector generalization remains untested for sectors not represented (energy, consumer staples, etc.). Fourth, defense overlays are runtime monkey-patches simulating deployment-realistic policies; production deployment of Provenance-Aware PM, Anomaly Filter, and Skeptic Agent would require engineering integration that we do not study. Fifth, our adversary model is non-adaptive: an attacker informed of the specific defense designs evaluated here could exploit each defense’s structural assumption to construct evasive payloads, in the spirit of Carlini-style adaptive attacks. We sketch the natural adaptive-attack surface against each defense:

- *Against Provenance-Aware PM*: fabricate named-source attribution by inventing plausible analyst names, advisory firms, or filing identifiers that pattern-match SEC EDGAR / 13F formats but do not correspond to existing entities. The citation-only variant would accept these without external lookup; the strict variant could still be defeated by constructing payloads with multiple syntactically independent pseudo-sources.
- *Against the FinBERT backend of the Anomaly Filter*: iteratively perturb payload phrasing to maximize cosine similarity to the real-news baseline centroid (a black-box gradient-free document-level optimization, analogous to embedding-space attacks against text classifiers). A payload tuned to score above the backend-specific filter threshold while preserving the attacker’s intended narrative would pass.
- *Against the Skeptic Agent*: construct payloads that simultaneously pass each of the seven checklist items by avoiding hedge vocabulary, distributing claims across two or three pseudo-independent fabricated sources, embedding numerical figures within plausible scale (e.g., contracts sized proportional to the target ticker’s revenue), and omitting price-reaction self-narration.

We do not test these adaptive variants in the present study. They constitute the natural next step in the evaluation arc this paper opens, and we expect that a serious adaptive-attacker analysis would quantitatively reduce all reported defense effects to some degree— how much, and whether composite defenses retain meaningful coverage under adaptive pressure, is the open empirical question.

A subtler limitation is the cross-ticker statistical sample size ($n = 5$). Cluster-bootstrap confidence intervals on five clusters are necessarily wide, and findings whose CIs straddle zero (e.g. cross-channel disinformation cross-ticker $\Delta = +0.04 [-0.12, +0.18]$) cannot be distinguished from null effects without more tickers. We treat such findings as suggestive rather than conclusive, and direct the reader’s attention primarily to those whose CIs exclude zero (bearish memory poisoning; Provenance-Aware PM and Skeptic Agent vs. fake news and cross-channel disinformation).

Scope of generalization across our findings. Within the results above, different findings carry different external-validity prospects, and we want to make this gradient explicit rather than treat the findings as uniformly generalizable.

The architectural memory access boundary (Section 5.3) is structural—defined by which agent reads the memory log, not by the agent LLM’s calibration—and is therefore the finding most likely to replicate on any LLM run through the same TRADINGAGENTS graph; we therefore treat it as the primary load-bearing claim of the paper. The direction-asymmetric bearish vulnerability (Section 5.4) is partly architectural and partly prior-driven: the architectural component (memory propagation under prior-aligned bearish narratives) should replicate, but the magnitude likely depends on the specific risk-aversion calibration of `gpt-4o-mini`. The high-prior backfire on NVDA (Section 5.1) is the most likely to be model-specific: it depends on the agent’s particular bullish prior on AI semis under `gpt-4o-mini`, which a different model may not share. The single-cell Skeptic-Agent backfire on bearish memory poisoning (Section 5.5) sits at the intersection of all three: single-ticker $N = 5$, single-LLM, single-direction. We therefore frame the architectural-attenuation finding as our primary contribution; the direction-asymmetric vulnerability and the defense-composition observations as secondary; and the per-ticker heterogeneity (NVDA backfire, BIIB sector mismatch) and the Skeptic-backfire as preliminary observations whose external validity awaits cross-LLM and cross-payload replication.

6.4 Future work

Three directions follow directly from the limitations above. First, *cross-LLM validation*: replicating the architectural-attenuation finding on `gpt-4o` and `Claude` would test whether the two-tier robustness is a property of TRADINGAGENTS’s architecture or of `gpt-4o-mini`’s priors. Second, *adaptive-attacker evaluation*: implementing the three adaptive-attack templates sketched in §6.3 (fabricated named sources against the provenance defense, embedding-space optimization against the FinBERT filter, checklist-aware evasion against the Skeptic Agent) would yield Carlini-style adversarial bounds on the defense matrix and is the most reviewer-anticipated extension. Third, *composite-defense engineering*: building and evaluating a deployment-grade pipeline that combines a citation-only provenance prompt, a FinBERT-backed tool-output filter, and a Skeptic Agent would test whether the mixed-attack leak is genuinely closable in practice or whether new attack classes emerge against the combined defense. Beyond these, extending the evaluation to a closed-loop backtest—feeding the linear ordinal-to-weight mapping of §4.5 into a portfolio simulator with transaction cost, drawdown tracking, and Sharpe-based aggregation along the lines of Lewen et al. [5]—would convert our exposure-level approximations into directly realized P&L, and adding a sector-matched fake news ablation (e.g. a Theranos-style biotech fake-clinical-trial seed) would test whether the BIIB and NVDA failure modes are payload-specific or sector-fundamental.

7 Conclusion

We measured 1,210 trials of adversarial information-channel manipulation against TRADINGAGENTS, an open multi-agent LLM trading framework, across five tickers and both bullish and bearish attack directions. Our central finding is that multi-agent debate provides *two-tier* adversarial robustness in the bullish direction: an architectural memory access boundary that, under the default TRADINGAGENTS configuration, attenuates pattern-matched memory poisoning at the analyst-stage perceptual layer in 49/50 trials (*perception-level attenuation*), and a multi-agent deliberation pipeline that absorbs cross-channel coordinated disinformation but does not act on it (*decision-level resistance*). Both tiers are partial: perception-level attenuation breaks when attack direction aligns with agent-prior bias (bearish memory poisoning succeeds at $\Delta = -0.42$, $p < 0.0001$, corresponding to a $\sim 21\%$ adverse portfolio-weight shift under a standard ordinal-to-exposure mapping), and decision-level resistance leaks on the Mixed Attack. We further report that adversarial robustness in this setting is not a property of any single defense layer but of *defense composition*: prompt-level (Provenance-Aware PM) and agent-level (Skeptic Agent) defenses each robustly reduce single-channel attacks across three tickers but leak on the Mixed Attack, while at the single-ticker $N = 5$ measurement on PLTR a tool-output stealth filter (Anomaly Filter with FinBERT backend) achieves the strongest single-cell mixed-attack defense ($\Delta = +0.80$). Together, these results suggest that for the configuration we test, the more informative unit of analysis for multi-agent LLM adversarial robustness may be the *composition* of architectural, policy, and filter-level defenses—each addressing a different failure mode the others leave open—rather than any single defense considered in isolation.

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